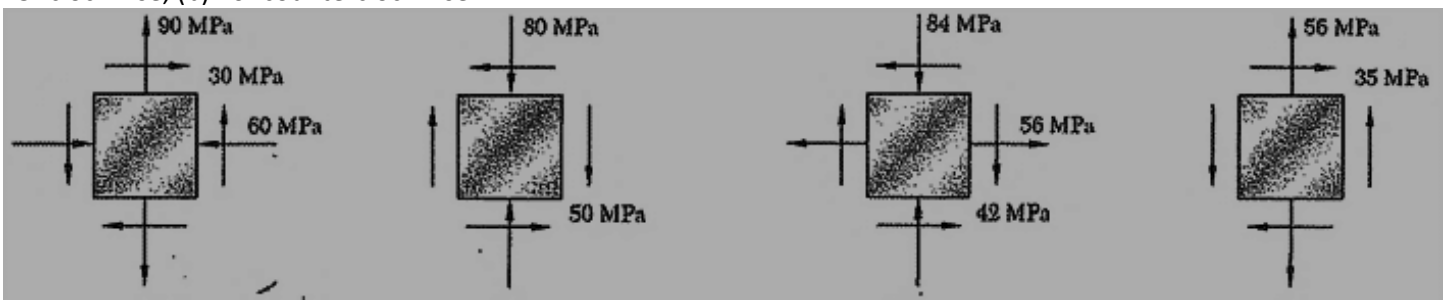
Tutorial Sheet **Mechanics** **Date: 23.05.2023**

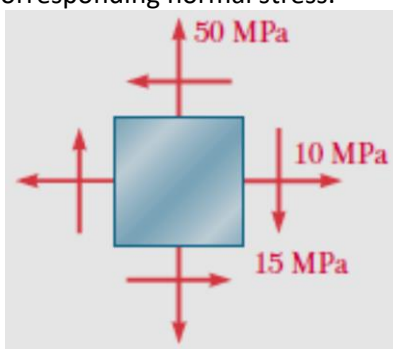
Prob. 1:

For the state of stress shown, determine the normal stress and shearing stresses after the element has been rotated (a) 25^0 clockwise, (b) 10^0 counterclockwise.



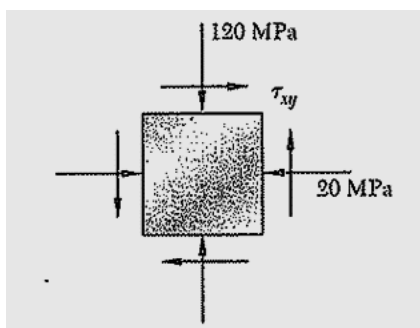
Prob. 2:

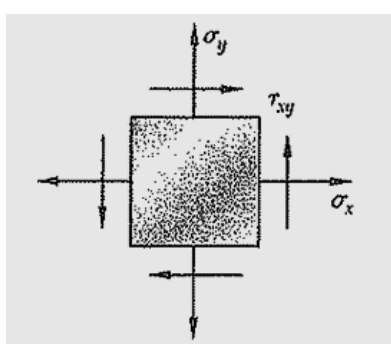
For the given state of stress, determine (a) the orientation of the planes of maximum in-plane shearing stress, (b) the maximum in-plane shearing stress, (c) the corresponding normal stress.



Prob. 3:

For the element shown determine the range of values τ_{xy} for which the maximum tensile stress is equal to or less than 60 MPa.



Prob. 4:

For the state of stress shown it is known that the normal and shear stress are directed as shown and that σ_x= 98MPa, σ_y= 63MPa and σ_{min}= 35 MPa. Determine (a) the orientation of the principal planes, (b) the principal stress σ_{max} , (c) the maximum in-plane shearing stress.